



Walmart Weekly Sales Business Analysis Report

From descriptive analysis to actionable retail insights

Introduction

Retail sales are influenced by multiple factors, including store performance, seasonality, holidays, environmental conditions, and economic conditions. Understanding these patterns can help businesses improve inventory planning, promotional strategies, and resource allocation.

This report analyzes Walmart's historical weekly sales data to identify important sales patterns across stores, time periods, holidays, environmental conditions, and economic indicators.

Purpose

The purpose of this analysis is to translate Walmart weekly sales data into evidence-based observations about:

- Store performance
- Sales seasonality and weekly trends
- Holiday demand
- Temperature and environmental conditions
- Economic indicators such as CPI, unemployment, and fuel prices

The analysis focuses on identifying meaningful patterns in the data and converting them into practical business insights and recommendations. The findings are then used to develop practical recommendations for inventory planning, promotions, and store-level decision-making."

Total Sales \$6.74B	Avg Weekly Sales \$1,046,965	Stores 45	Holiday Uplift 7.8%
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1. Executive Summary

The analysis indicates that store-level differences and seasonality are among the strongest observable patterns in weekly sales. A relatively small group of stores contributes a disproportionate share of revenue, while November and December show the strongest average weekly sales. Holiday weeks show higher average sales than non-holiday weeks.

Economic indicators are less decisive than the initial interpretation suggests. Unemployment has a weak negative association with sales, while CPI has an even weaker negative association. Fuel price has almost no linear relationship with weekly sales in this dataset.

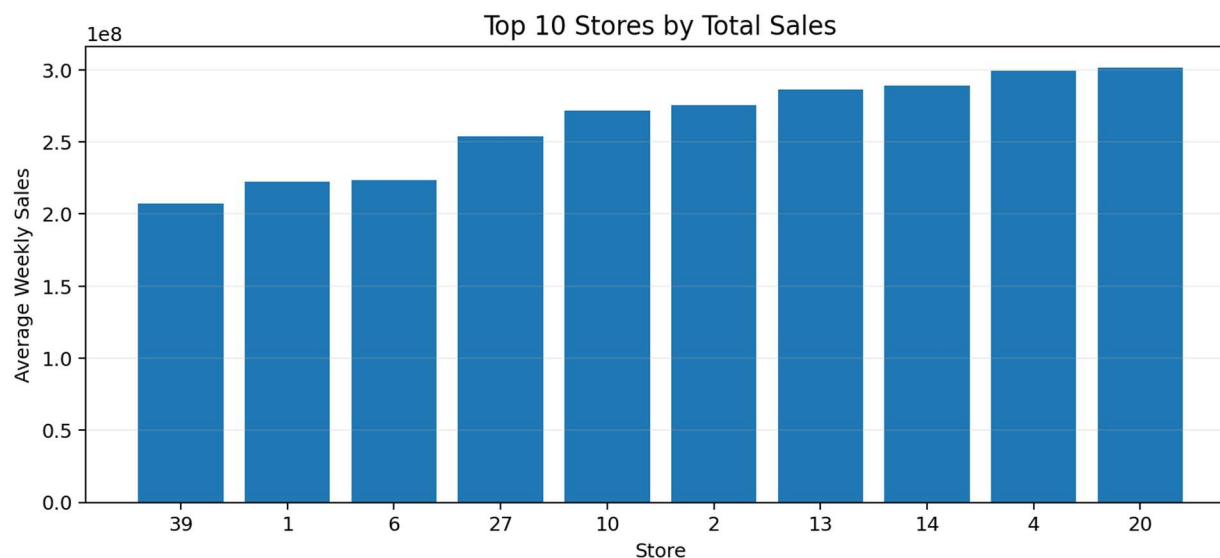
Temperature appears nonlinear: sales are weaker at extreme temperatures and more stable in moderate ranges.

Key Business Takeaways

- Prioritize inventory availability and replenishment capacity at consistently high-performing stores.
- Prepare additional inventory, staffing and promotional capacity for November and December.
- Use holiday promotions selectively and measure incremental sales and margin.
- Treat extreme temperatures as a potential demand-planning signal.
- Treat CPI, unemployment, and fuel price as supporting indicators rather than standalone explanations of sales.

2. Store Performance

Store performance is highly uneven. The top 10 stores contribute approximately 39.1% of total sales, This suggests that a store-segmented strategy may be more effective than applying the same approach across all locations.

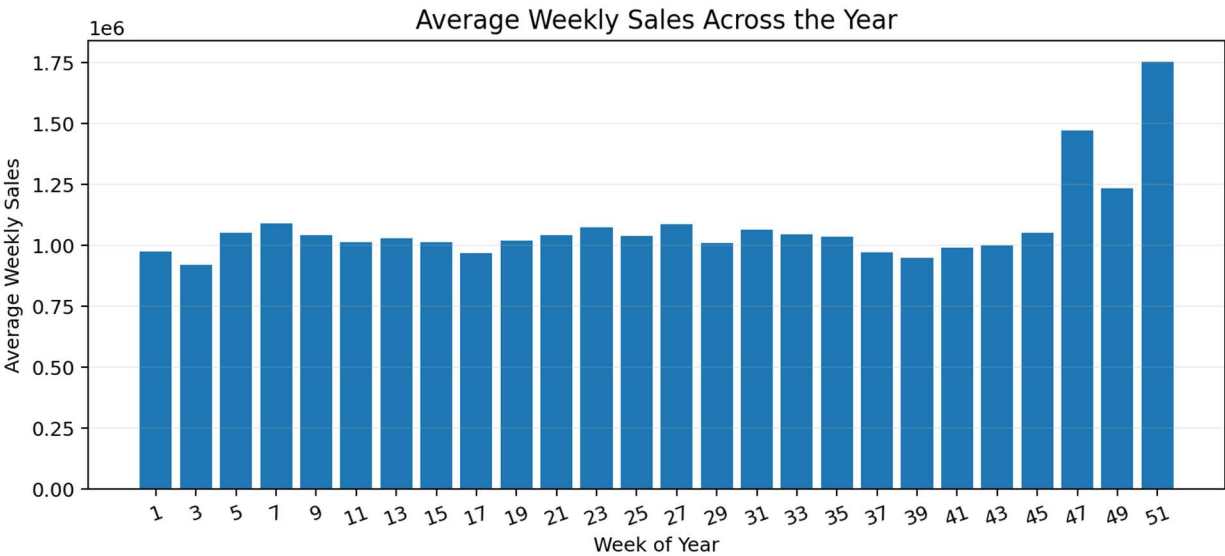
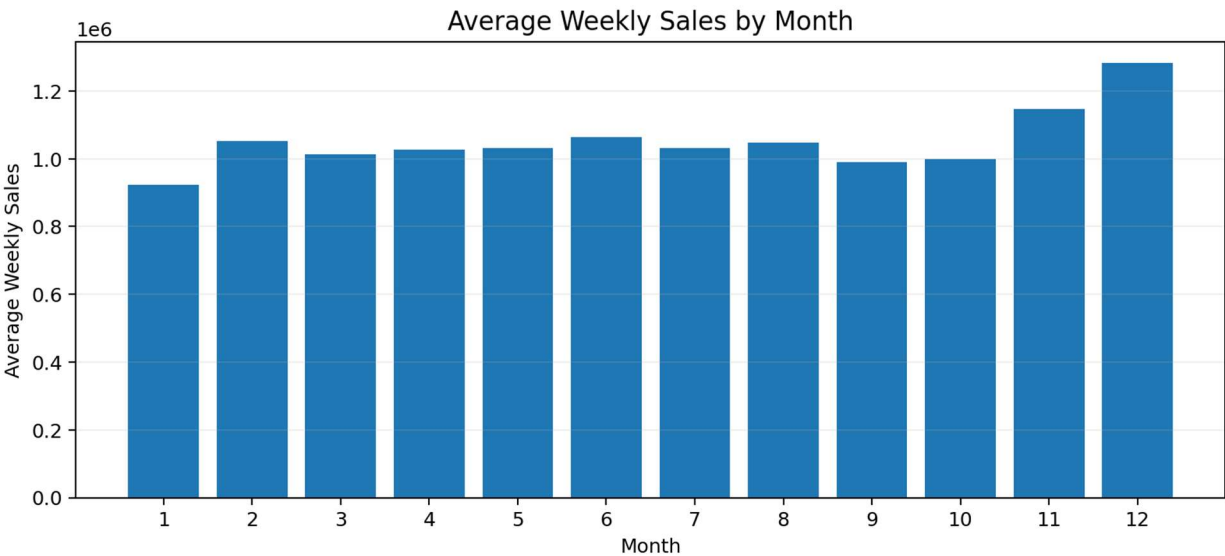


Business implication: high-performing stores should receive stronger inventory protection and capacity planning during peak periods. Promotions should be evaluated using incremental sales and profitability, not simply revenue volume.

3. Seasonality and Week-of-Year

Sales strengthen materially toward year-end. November averages approximately \$1,147,266 per week and December approximately \$1,281,864, compared with an overall weekly average of \$1,046,965.

Weekly patterns also vary throughout the year, suggesting that the timing of a sales week is an important component of demand.



4. Holiday Effect

Holiday weeks average \$1,122,888 in weekly sales compared with \$1,041,256 during non-holiday weeks, representing an uplift of approximately 7.8%.

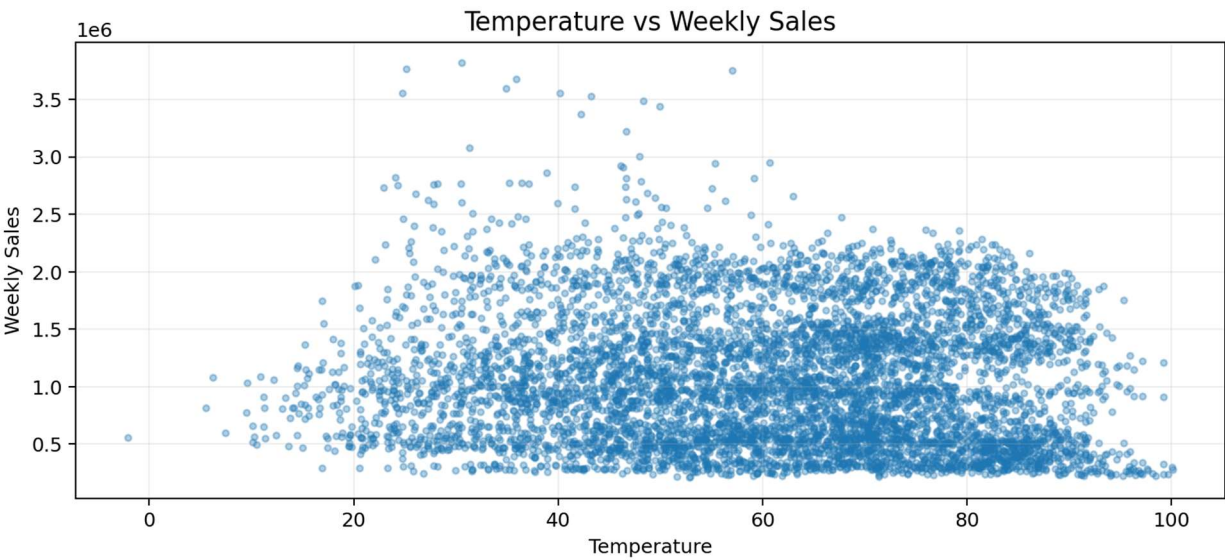
Period	Average Weekly Sales	Comparison
Holiday	\$1,122,888	Baseline + uplift
Non-holiday	\$1,041,256	Baseline

This suggests that holiday campaigns may be a useful opportunity, but their effectiveness should be evaluated using incremental sales, margin, and inventory costs, but the business should evaluate incremental sales, gross margin and inventory costs before scaling promotions.

5. Sales Drivers and External Factors

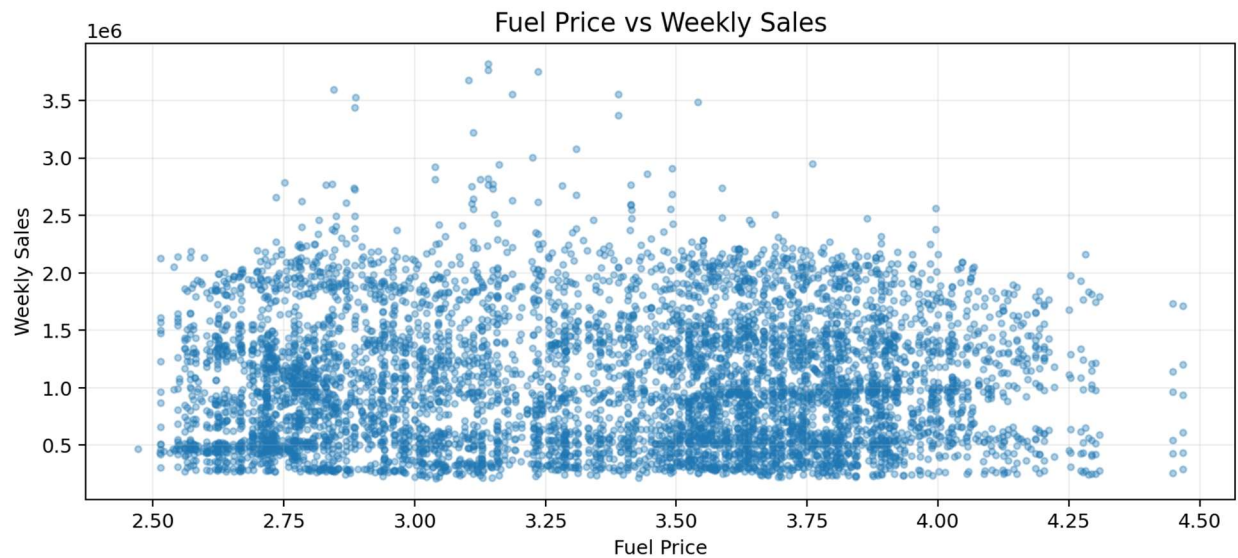
Temperature

Temperature shows a nonlinear relationship with sales. Sales tend to be weaker at extreme temperatures, while moderate temperature ranges show relatively more consistent performance. This makes temperature a potentially useful supporting variable for demand planning.



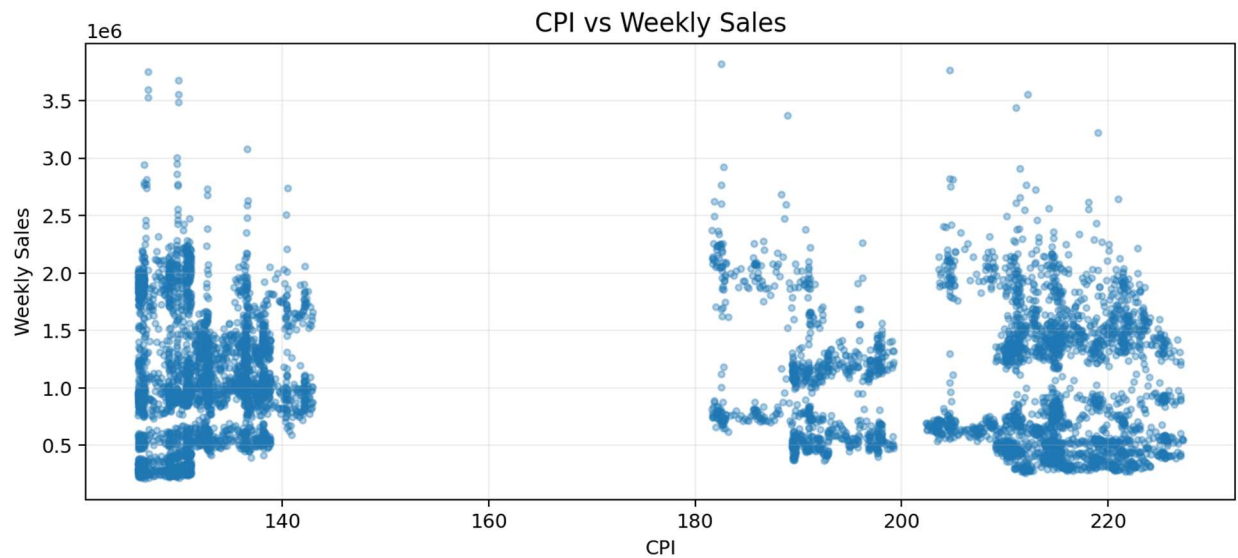
Fuel Price

Fuel price shows almost no linear relationship with weekly sales, with a correlation of approximately 0.009. The data therefore does not support my initial assumption that sales drop sharply once fuel prices exceed 4.



CPI

CPI has a weak negative correlation with weekly sales of approximately -0.073. This suggests some association, but the relationship is too weak to consider CPI a major standalone sales driver.



Unemployment

Unemployment has a weak negative correlation with weekly sales of approximately -0.106. This is directionally consistent with the possibility of lower consumer spending during periods of higher unemployment, although the dataset cannot establish a causal relationship.



6. Key Findings from Analysis

What I Observed	What I Found	My Interpretation
Week affects sales	Sales vary across different weeks, with stronger performance toward the end of the year	Week/seasonality appears to have an important effect, although I cannot confirm that salary cycles are the reason
Extreme temperatures affect sales	Sales tend to be lower at very low and very high temperatures	Extreme weather may negatively affect customer demand
Fuel price affects sales	I initially expected sales to fall sharply above 4, but the data does not show a consistent sharp decline	Fuel price does not appear to be a major factor in this dataset
CPI affects sales	CPI has a weak negative relationship with sales	CPI may have some influence, but the relationship is not strong enough to consider it a key driver
Unemployment affects sales	Sales tend to be lower when unemployment is higher	This is consistent with the idea that weaker employment conditions can reduce consumer spending, but the data alone cannot prove causation
Holidays affect sales	Holiday weeks have approximately 7.8% higher average sales	Holidays appear to provide a positive sales opportunity
November and December perform strongly	These months have the highest average weekly sales	Year-end seasonality should be considered when planning inventory and promotions
A small number of stores perform exceptionally well	The top 10 stores contribute approximately 39% of total sales	High-performing stores should receive greater focus on inventory, promotions, and campaigns

7. Recommended Business Strategy

1. Store segmentation

Create tiers based on sales volume and consistency. Allocate inventory, staffing and promotional budgets according to store potential.

2. Peak-season planning

Increase inventory readiness and operational capacity ahead of November and December.

3. Targeted promotions

Use holiday and seasonal offers as controlled experiments. Track incremental sales, margin and stock-out rates.

4. Demand planning

Incorporate temperature and seasonal variables into forecasting rather than assuming a linear relationship.

5. Data-driven testing

Test week-based promotional strategies using A/B testing or historical matched-store comparisons before rolling them out.

8. Limitations and Next Steps

This is observational analysis. Correlation does not imply causation, and store, time and geography can confound relationships. The dataset does not contain customer salary dates, transaction-level customer behavior, promotion details, inventory, margin or stock-out information.

- Add promotion and inventory data to test whether offers actually drive incremental sales.
- Use store-level and time-controlled regression to separate store effects from economic effects.
- Test lagged economic variables and seasonal interactions.
- Use A/B testing or quasi-experimental methods to evaluate proposed promotions.
- Compare model predictions with business rules and quantify the value of forecast-driven inventory planning.

9. Conclusion

The analysis suggests that store-level performance and seasonality are the most actionable patterns in the dataset. Sales increase significantly toward the end of the year, and holiday weeks also show higher average sales. Temperature, unemployment, and CPI provide additional context, while fuel price does not show a meaningful linear relationship with sales.

Based on these findings, I would focus on store-level segmentation, peak-season inventory planning, targeted promotions, and further testing of week-based sales strategies. Additional data on promotions, inventory, margins, and customer behaviour would allow these observations to be tested more rigorously.